

CIS 6270

Module 4: Discrete Diffusion

Lecture 4.2: Continuous-Time Markov Chains

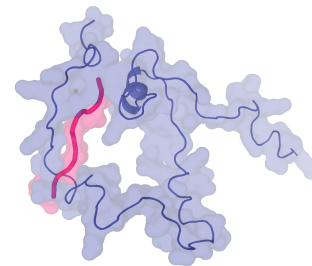
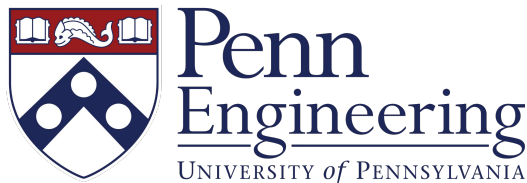
Tue, Oct 6, 2026

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MDLM, D3PM, uniform diffusion, SEDD, block diffusion, and simplex diffusion are each one choice inside this framework.

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The transition probability $P_{s,t}(x, y)$ is the probability that the process is at state y at time t , given that it was at state x at time s .

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The diagonal entry is minus the total rate of leaving x . We will check these two rules for every model in this lecture.

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The four probabilities sum to one, and when A changes, C is twice as likely as G or T.

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For A, the total rate is 4, so the step must satisfy $h \leq 0.25$. We will need the same check when we sample.

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When A jumps, we draw the destination in proportion to the rates:

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A waiting time and a destination describe the whole process

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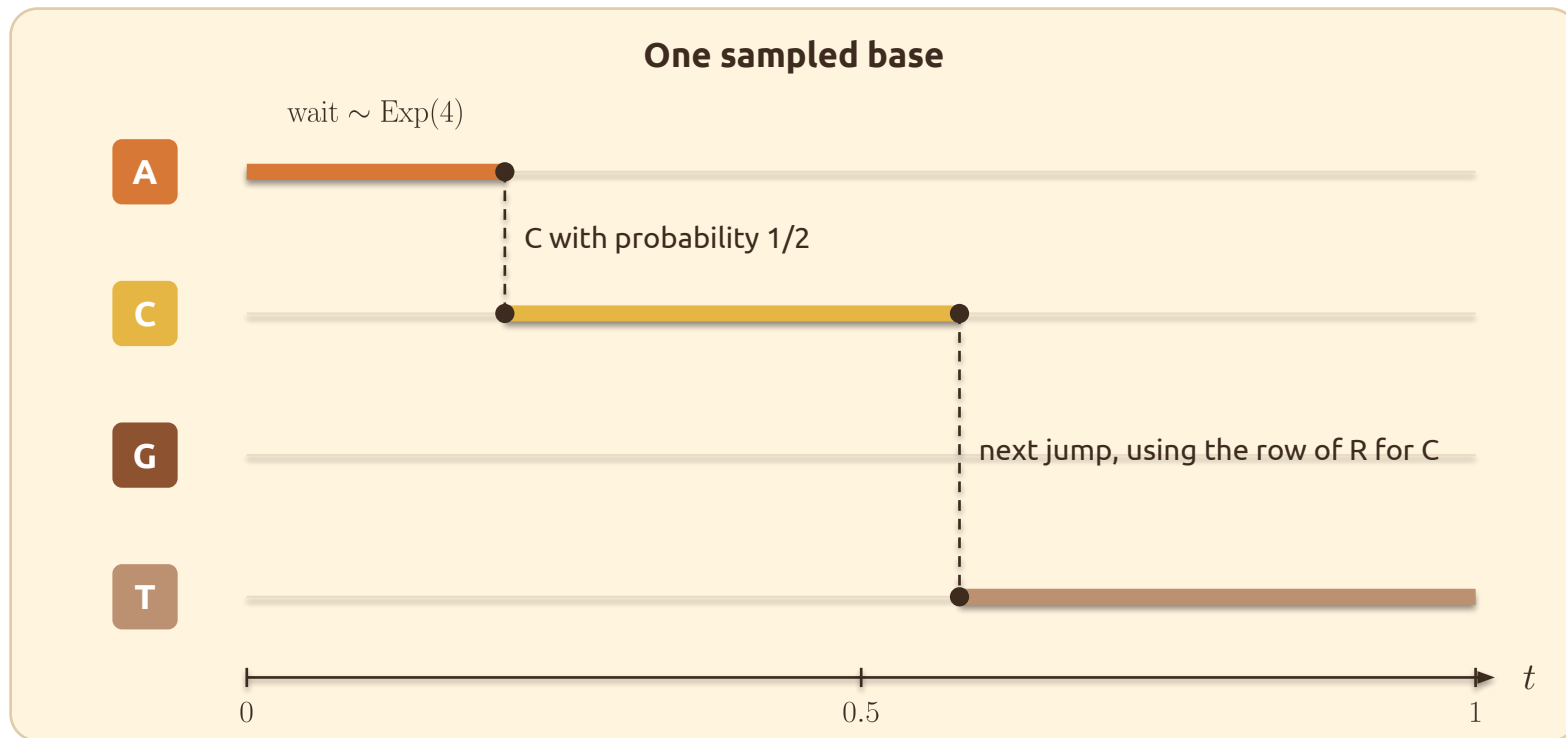
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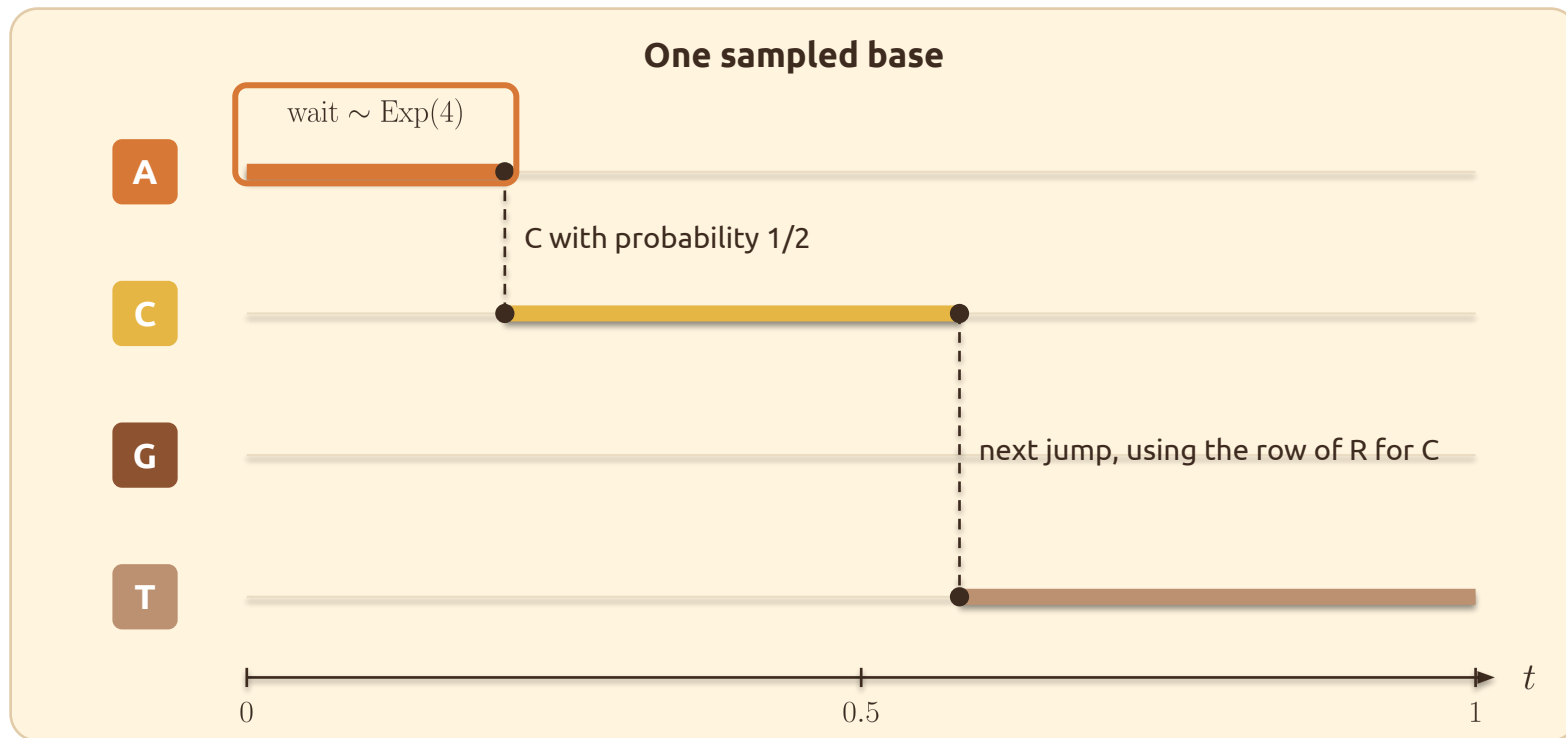
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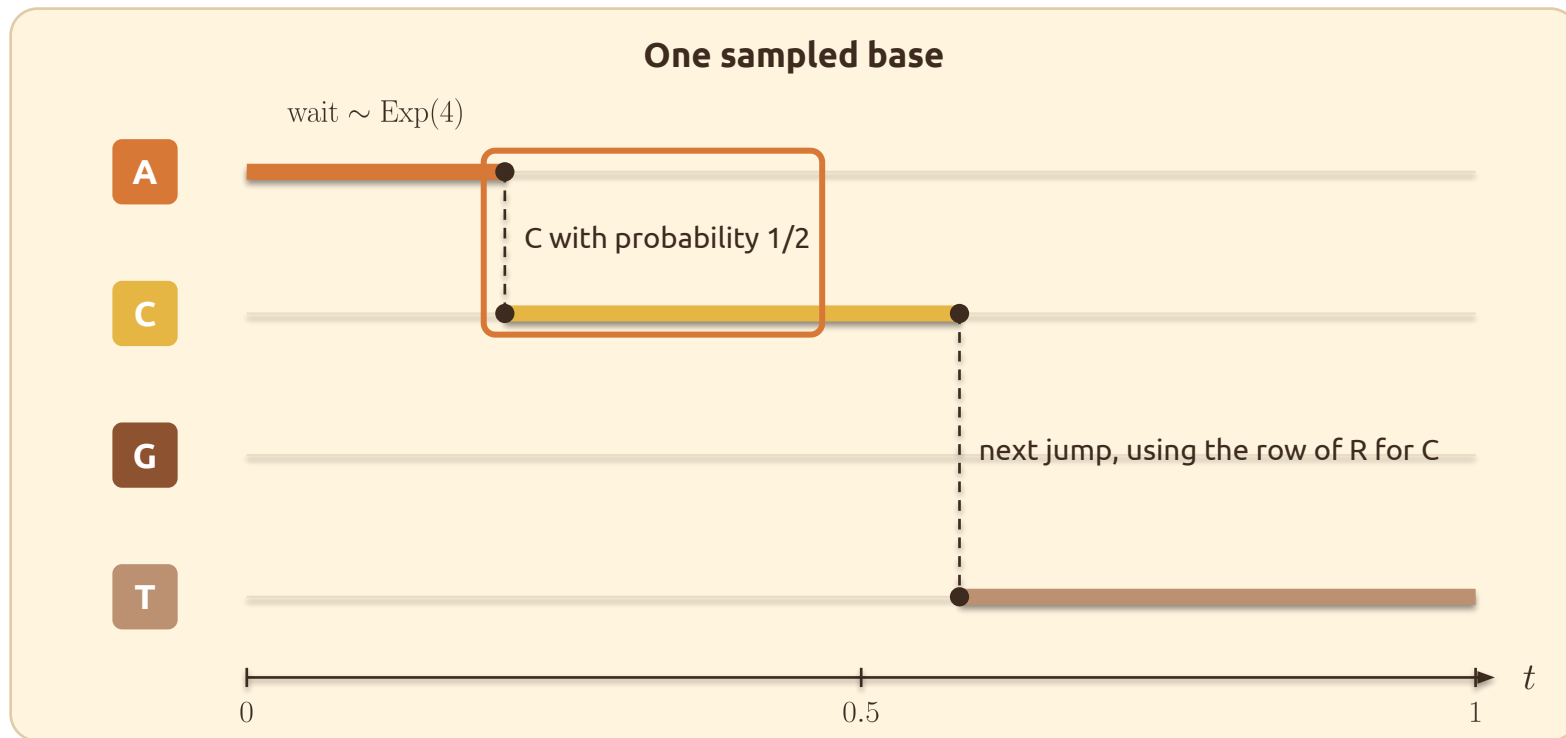
Follow one sampled base from $t = 0$ to $t = 1$



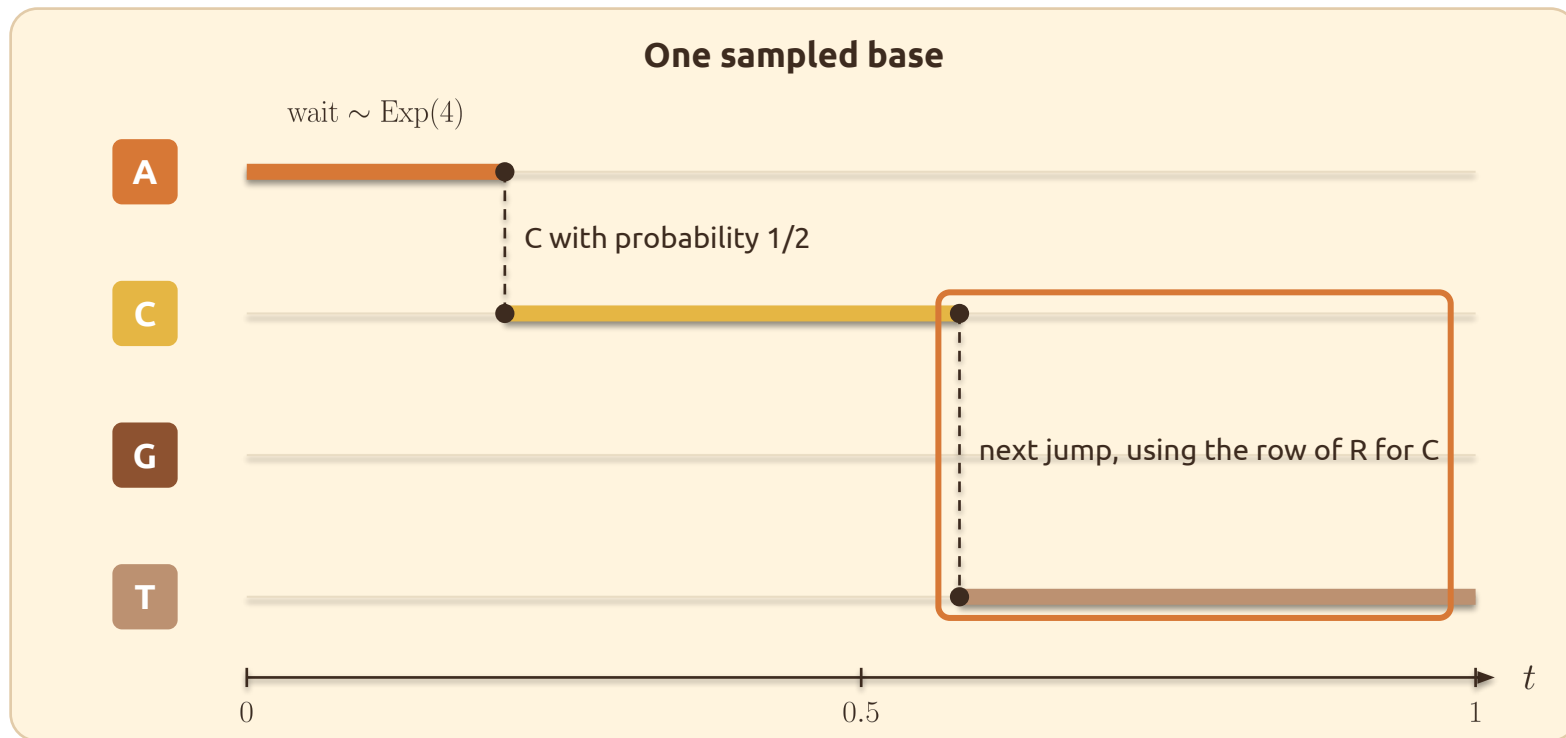
It waits an exponential time with rate 4, the total rate of leaving A



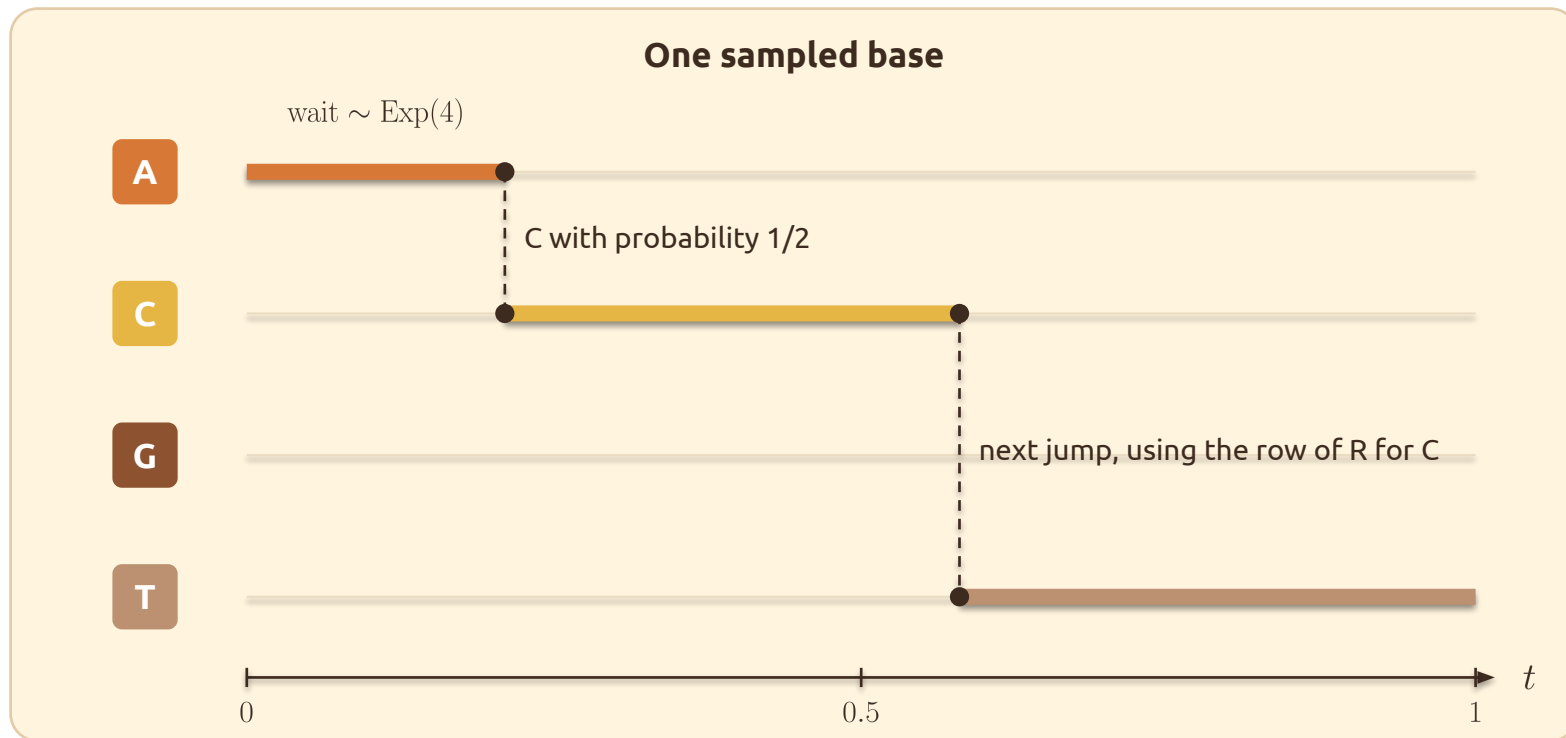
When it jumps, we draw the destination in proportion to the rates, and C has probability $1/2$



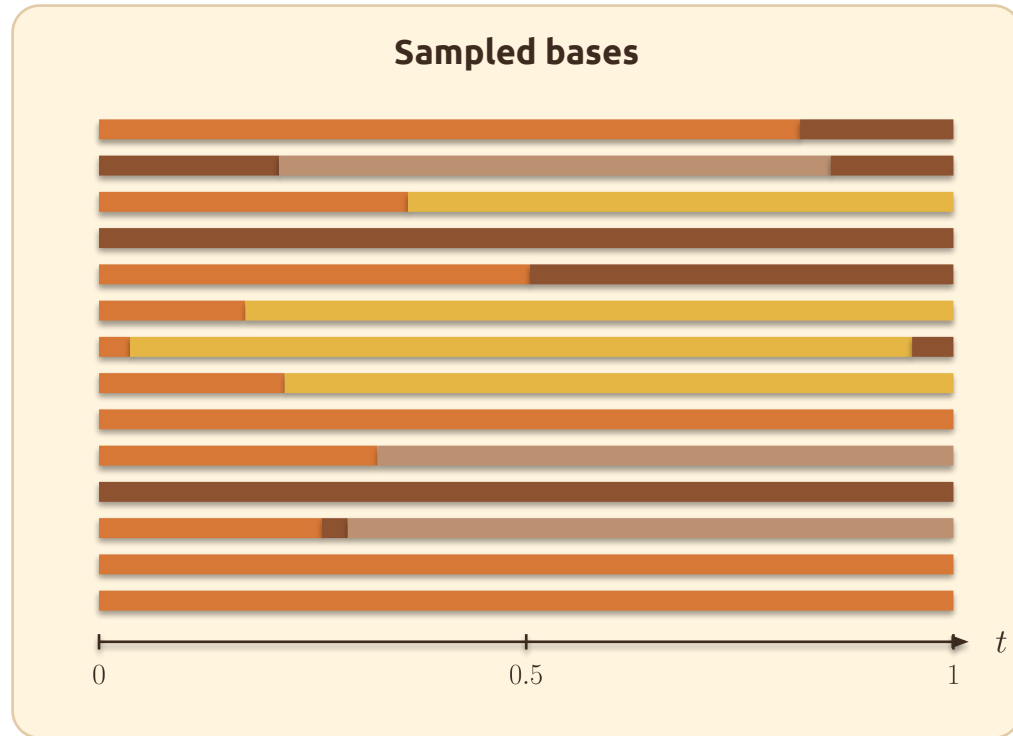
From C, the next waiting time and destination depend on the row of R for C



Now find the distribution of a path of waits and jumps

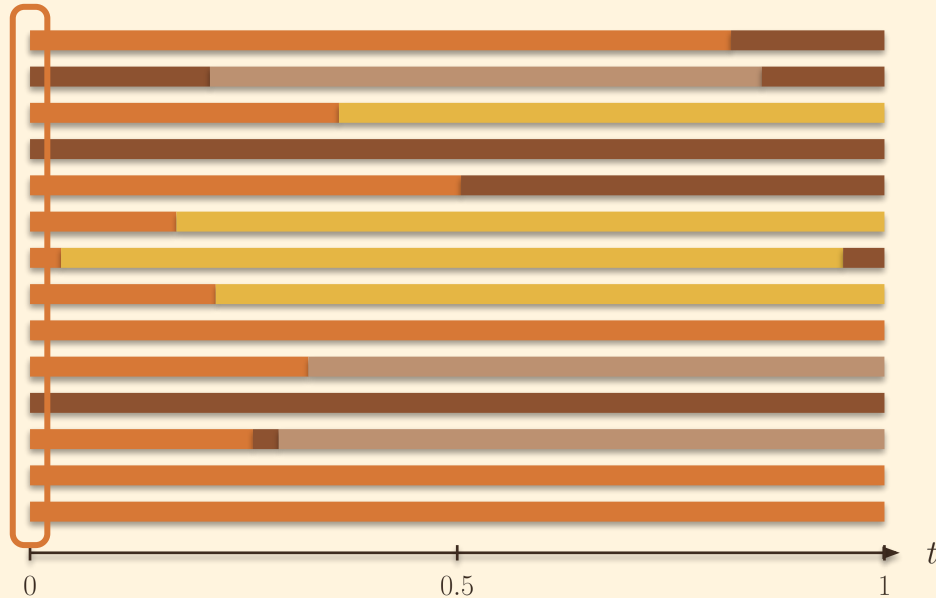


Fourteen bases, started mostly at A, follow their own paths

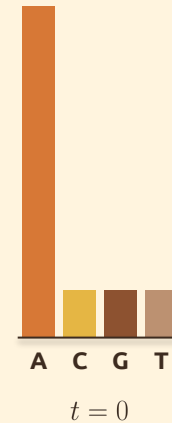


At $t = 0$, seven of every ten samples sit at A

Sampled bases

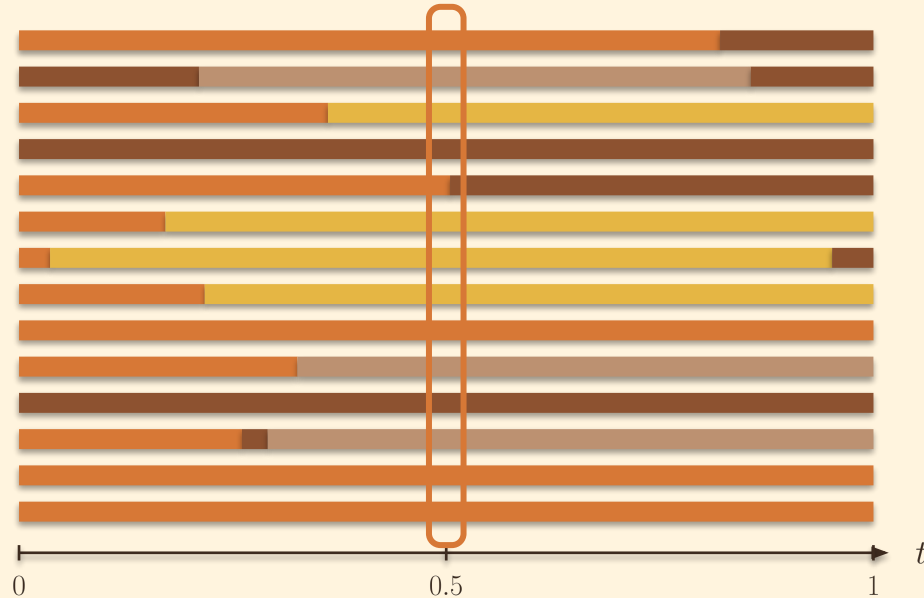


Proportions

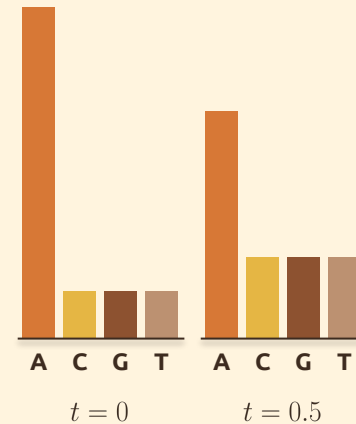


Collect every sample at one time to get the proportions in that time's bars

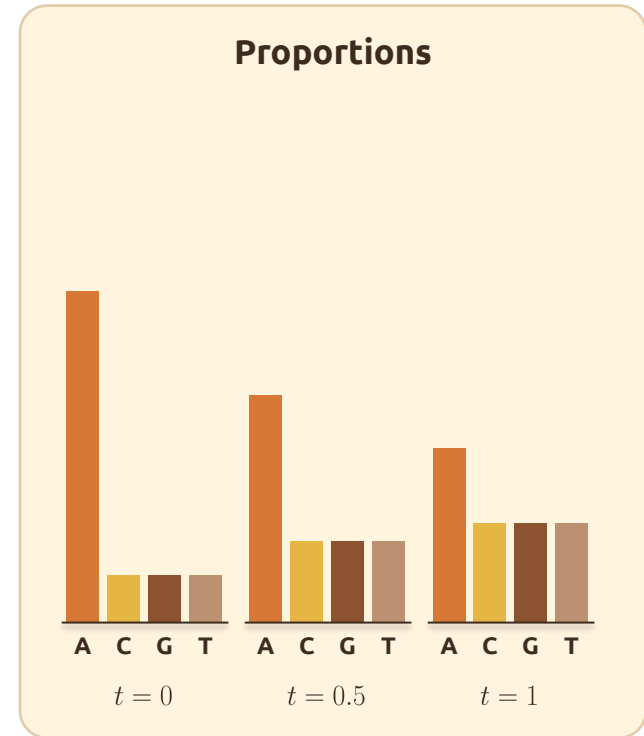
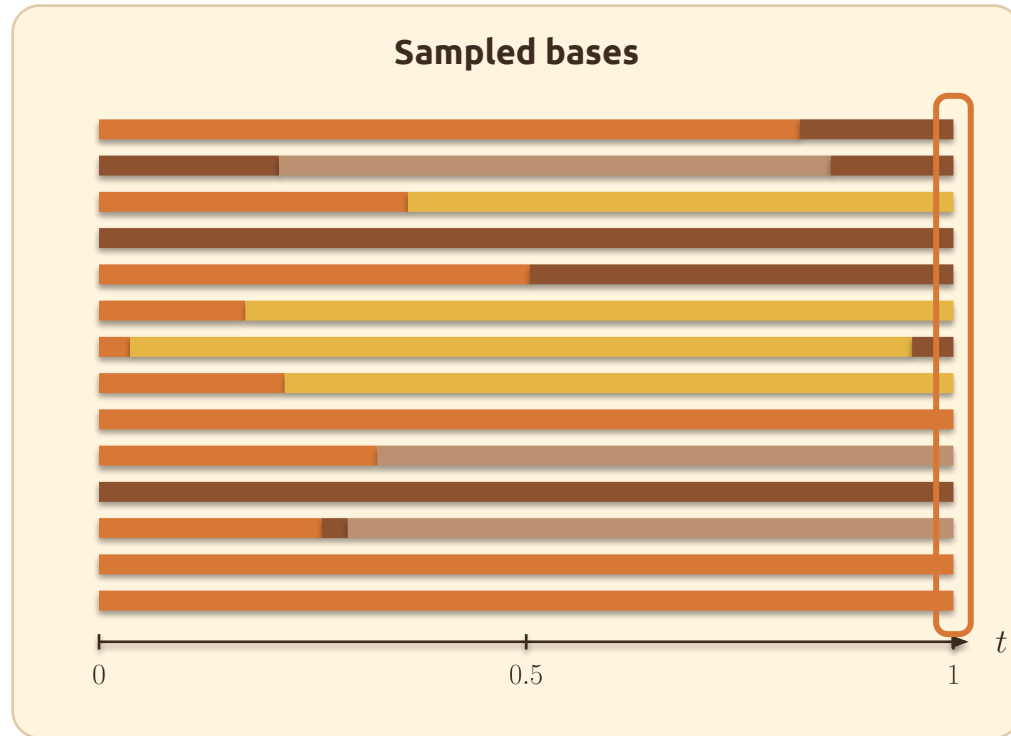
Sampled bases



Proportions

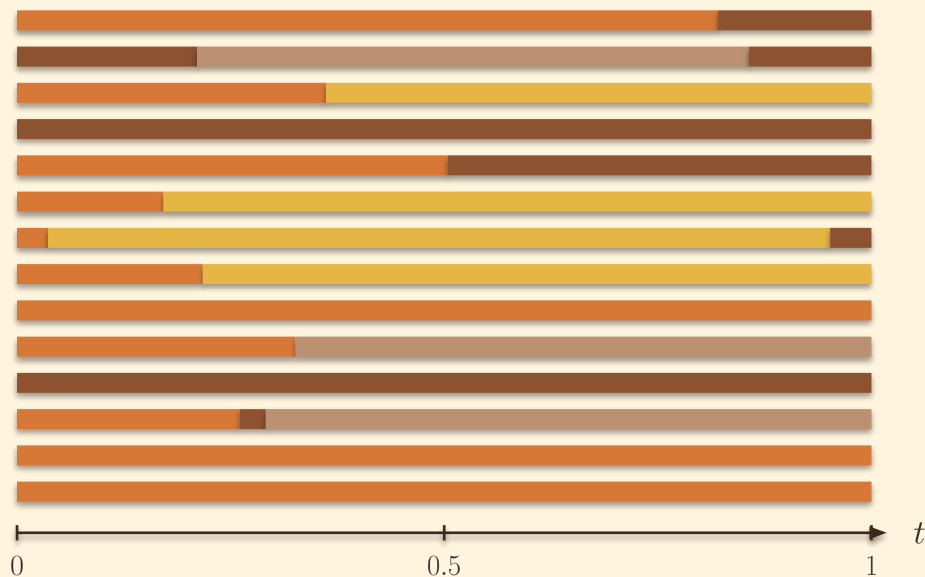


By $t = 1$, the four bases are nearly equally common

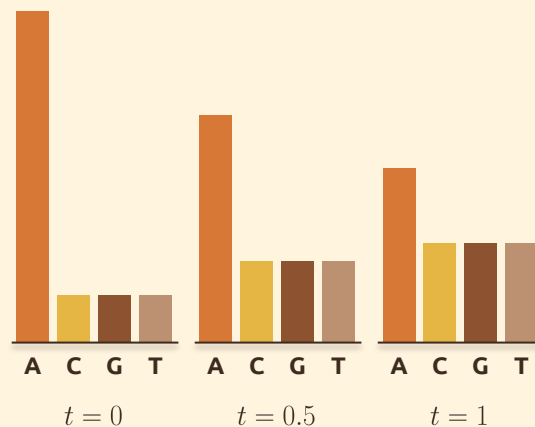


Individual samples jump, and their proportions change smoothly

Sampled bases

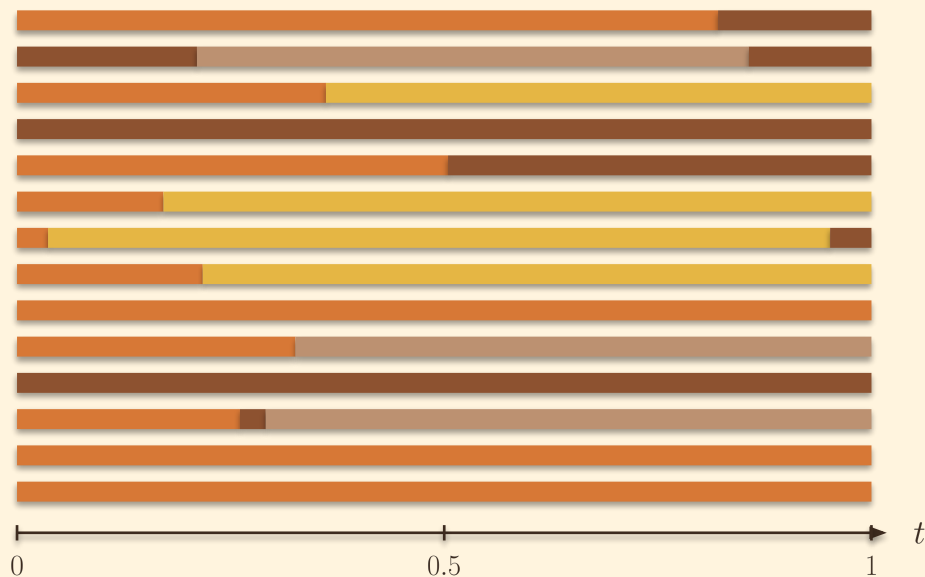


Proportions

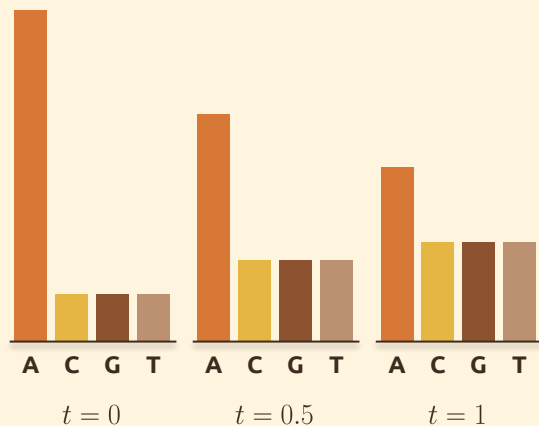


We need an equation for how those proportions change

Sampled bases



Proportions



Recall the probability flux from Lecture 2

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No flux leaves a state with no probability, whatever its rates, and no flux goes to y from a state with no rate toward y , whatever its probability.

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Both factors matter, so let's add up the flux at one state.

Now write the probability of y after a short interval

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After a short interval, the probability of y is a sum over every current state. Substitute the short-interval expansion:

$$p_{t+h}(y) = \sum_x p_t(x) P_{t,t+h}(x, y) = p_t(y) + h \sum_x p_t(x) R_t(x, y) + o(h)$$

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The probability $p_{t+h}(y)$ after a short interval h equals the sum over states x of $p_t(x)$ times the transition probability $P_{t,t+h}(x, y)$, which is $p_t(y)$ plus h times the sum over x of $p_t(x)$ times $R_t(x, y)$, up to $o(h)$.

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Subtract $p_t(y)$, divide by h, and let h go to zero:

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This is the Kolmogorov forward equation. For jump processes, we call it the master equation.

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The rate of change of $p_t(y)$ equals the sum over every other state x of $p_t(x)$ times $R_t(x, y)$, the flux arriving at y , minus the sum over every other state x of $p_t(y)$ times $R_t(y, x)$, the flux leaving y .

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In Lecture 2, density changed by the net flux through the boundary of a region. Here it changes by the net flux along the edges into and out of a state.

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Sum the master equation over every state y :

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Each flux leaves one state and enters another, so it appears once with a plus sign and once with a minus sign.

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Under the master equation, probability moves between states, and none of it is created or destroyed.

Let's evolve four DNA probabilities

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The probabilities p_t of A, C, G, and T start at 0.1, 0.2, 0.3, and 0.4.

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$$\dot{p}_t(A) = (0.2 + 0.3 + 0.4)(1) - (0.1)(3) = 0.6$$

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The rate of change $\dot{p}_t(A)$ equals the probability $0.2 + 0.3 + 0.4$ at the other bases times the rate 1 into A, minus the probability 0.1 at A times its total rate 3 of leaving, which is 0.6.

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$$\dot{p}_t(y) = 1 - 4p_t(y), \quad \dot{p}_t = (0.6, 0.2, -0.2, -0.6)$$

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For every base y , the rate of change $\dot{p}_t(y)$ equals one minus 4 times $p_t(y)$, which is 0.6, 0.2, -0.2 , and -0.6 for A, C, G, and T.

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The changes sum to zero. Rare bases gain probability, and common bases lose it.

Every base moves toward one quarter, and the total stays one

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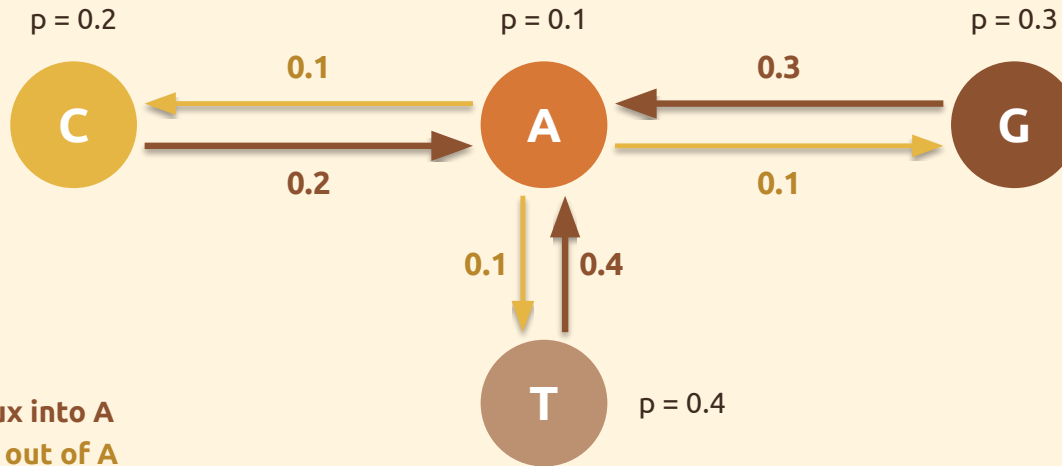
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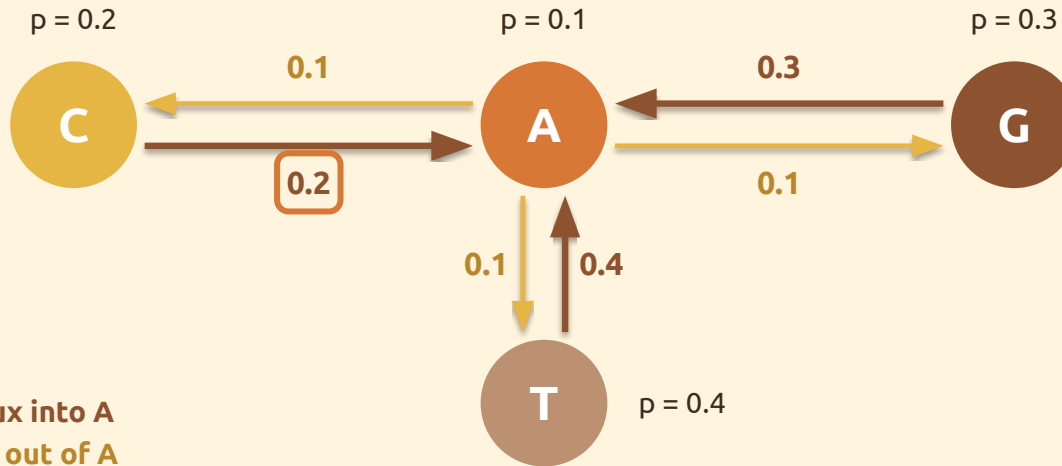
Let's see that balance at A

$$\dot{p}_t(A) = \underbrace{0.2 + 0.3 + 0.4}_{\text{flux in}} - \underbrace{0.1 \times 3}_{\text{flux out}} = 0.6$$

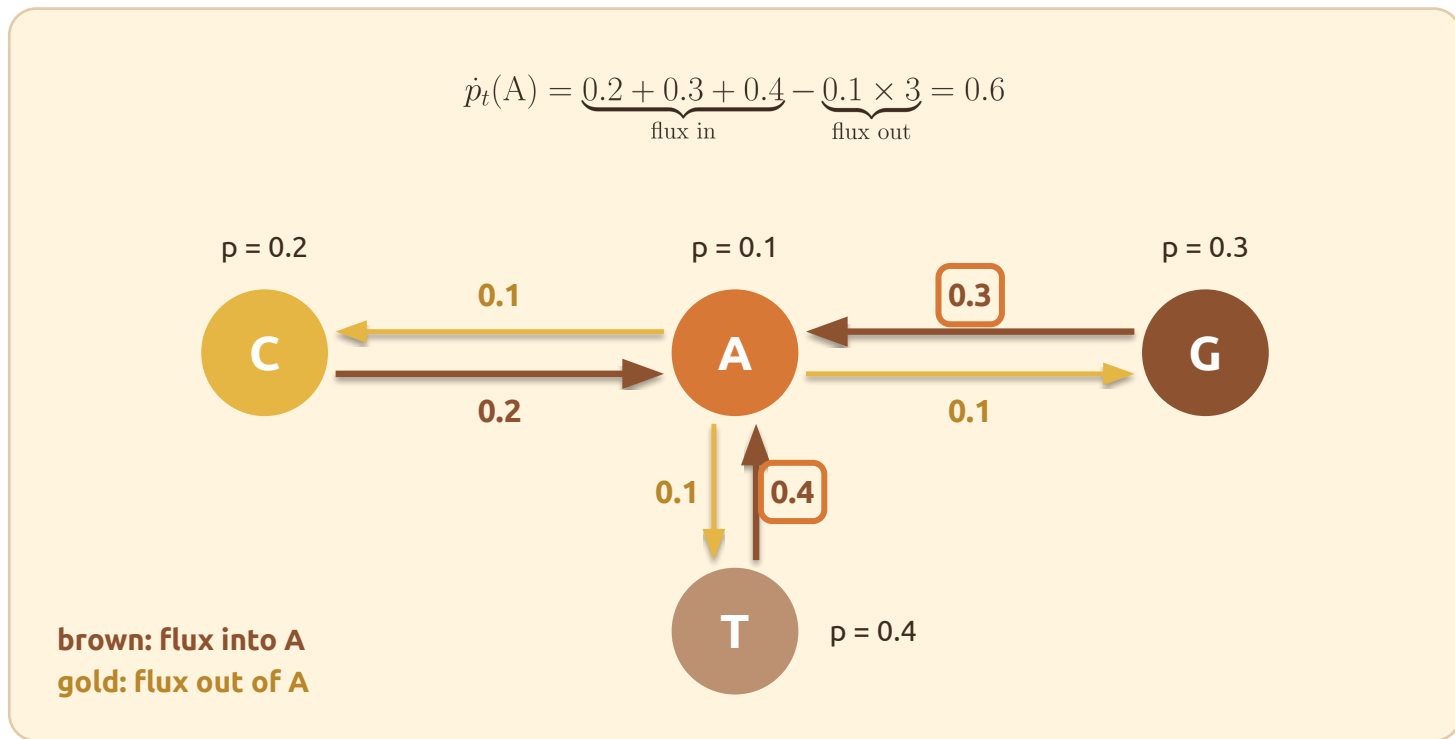


Probability 0.2 arrives from C at rate 1

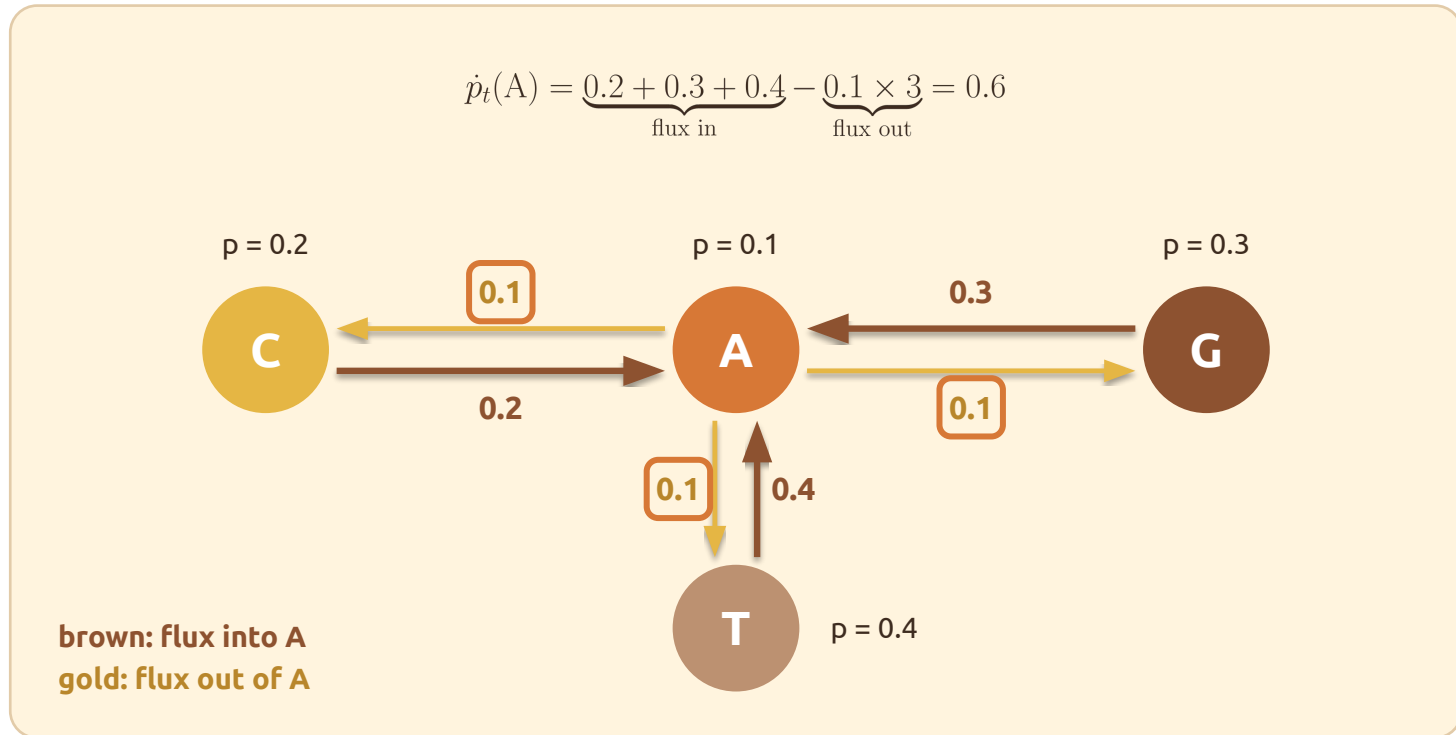
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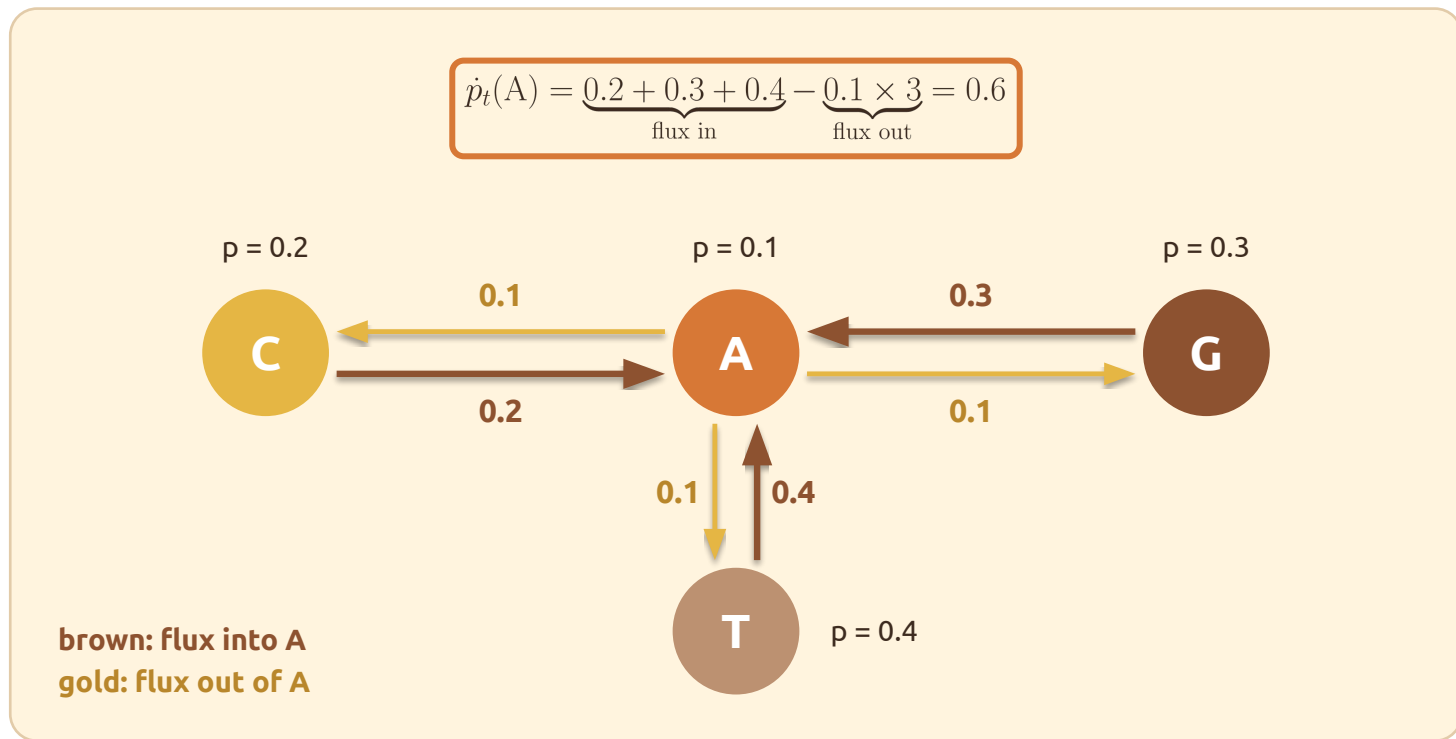
Probability 0.3 arrives from G, and 0.4 from T, at the same rate



Probability 0.1 leaves A for each of the other three bases

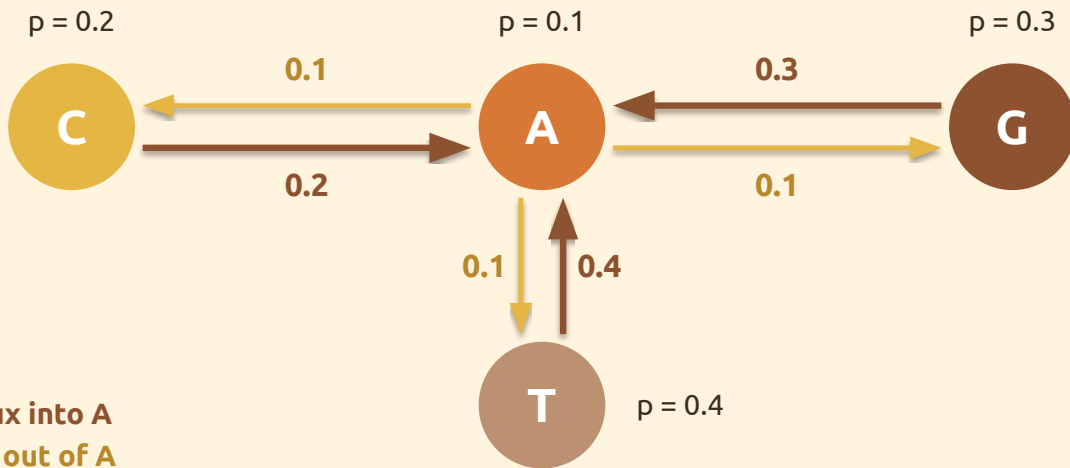


Probability 0.9 flows into A while 0.3 flows out, so $p_t(A)$ grows at rate 0.6



Solving this equation by hand at every time would be slow

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$$p_t = p_0 e^{tR}, \quad e^{tR} = \sum_{k=0}^{\infty} \frac{t^k R^k}{k!}$$

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The probability vector p_t equals the initial vector p_0 times the matrix exponential e^{tR} , which is the sum over k from zero to infinity of t^k times R^k divided by k factorial.

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So the transition matrix over an interval of length t is $P_{0,t} = e^{tR}$

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The matrix exponential e^{tR} times the all-ones vector $\mathbf{1}$ equals $\mathbf{1}$ plus the sum over $k \geq 1$ of $t^k R^k \mathbf{1}$ divided by k factorial, and every $R^k \mathbf{1}$ is zero, so every row of e^{tR} sums to one.

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Choose λ at least as large as every total outgoing rate, so that $R + \lambda I$ has no negative entries:

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The transition probability $P_{s,u}(x, z)$ from x at time s to z at time u equals the sum over every intermediate state y of $P_{s,t}(x, y)$, the probability of reaching y at time t , times $P_{t,u}(y, z)$, the probability of then reaching z at time u .

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To survive until u , a token first had to survive until t , and this is the argument we used for the masking kernel.

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The rate matrix R_t at time t equals the rate $\beta(t)$ times the fixed rate matrix R , and the clock $\tau(t)$ equals the integral of $\beta(s)$ over times s from 0 to t .

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With the rate matrix R , we choose where probability can move, and with the schedule $\beta(t)$, how fast it moves.

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The rate matrix R_{π} equals the all-ones column $\mathbf{1}$ times the row $\pi^{\top}$, a matrix whose rows all equal the distribution π , minus the identity matrix I .

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A matrix with $M^2 = M$ is called idempotent, and every power M^k equals M .

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The matrix exponential of τ times $M - I$ equals $e^{-\tau}$ times the matrix exponential $e^{\tau M}$, because the identity commutes with M .

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Every power M^k equals M , so the series collapses:

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Every row of this matrix is a mixture of staying put and a fresh draw from π .

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Write $\alpha_t = e^{-\tau(t)}$ for the survival probability. Each row of the transition matrix reads

$$q_t(x \mid x_0) = \alpha_t \delta(x, x_0) + (1 - \alpha_t) \pi(x)$$

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The probability $q_t(x \mid x_0)$ of the noisy token x given the clean token x_0 equals the survival probability α_t times $\delta(x, x_0)$, which is one when x equals x_0 and zero otherwise, plus one minus α_t times $\pi(x)$, the probability that π assigns to x .

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Every survival schedule α_t we pick corresponds to exactly one rate function $\beta(t)$.

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With π equal to δ_m , which has all of its probability on the mask token m , the probability $q_t(x \mid x_0)$ equals α_t times $\delta(x, x_0)$ plus one minus α_t times $\delta(x, m)$, which is one only at the mask.

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With masking, we erase the identity of a token, and with uniform replacement, we hide it among the other tokens.

From one formula, we get two models, depending on π

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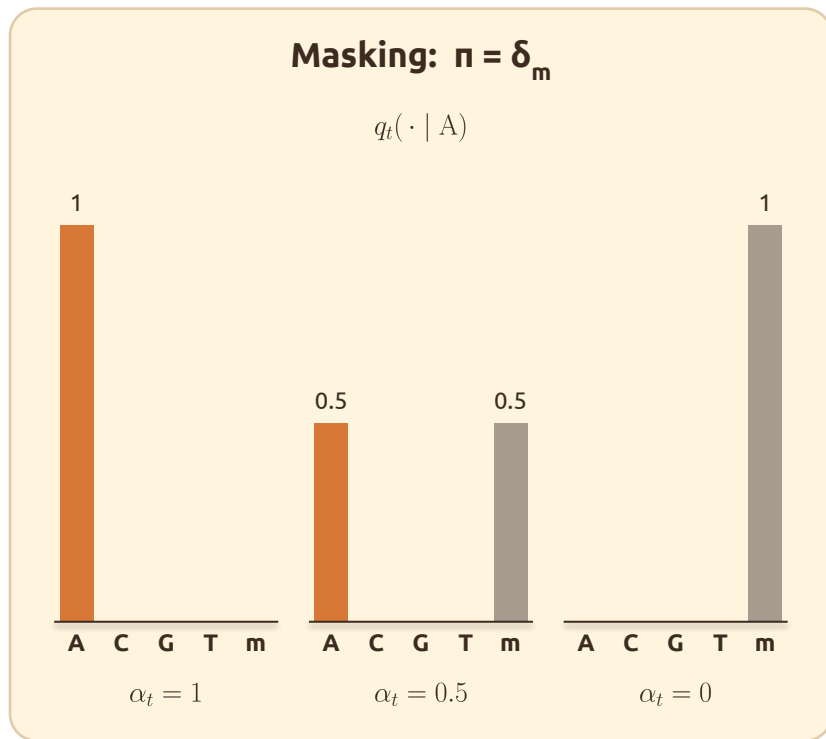
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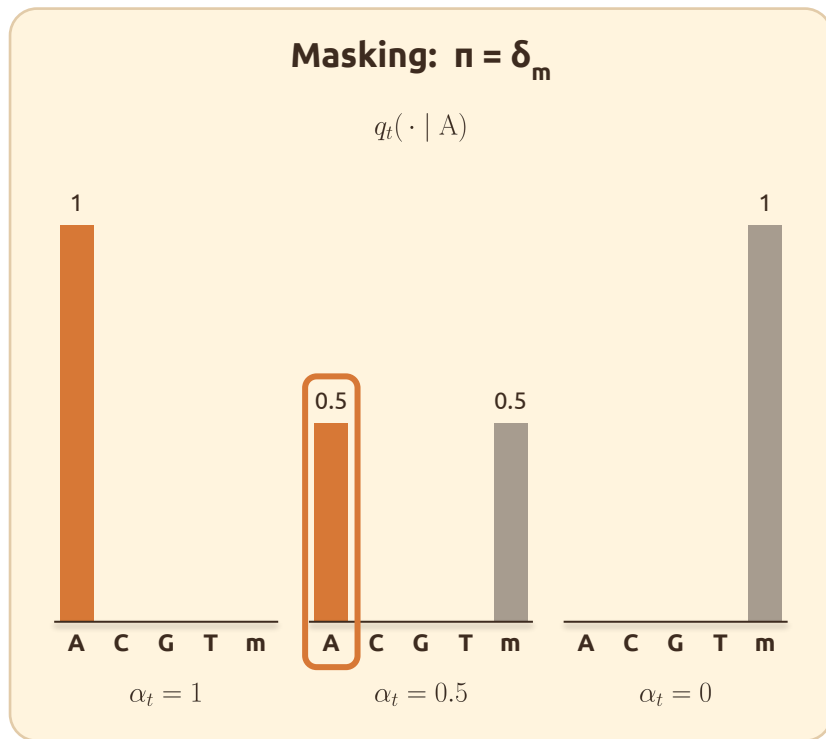
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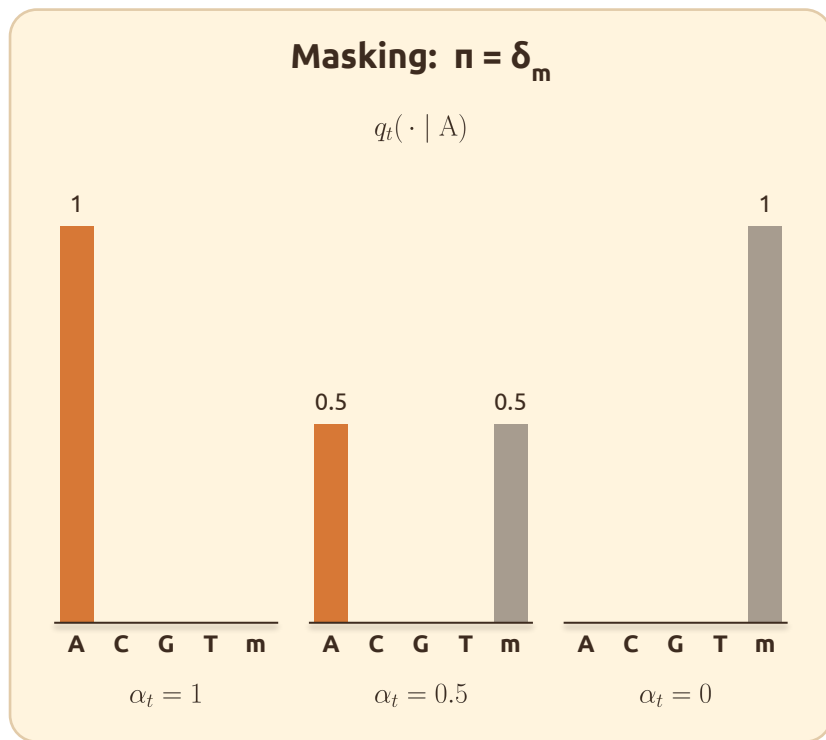
Let's see both kernels at three noise levels



Under masking, the clean base stays with probability α_t



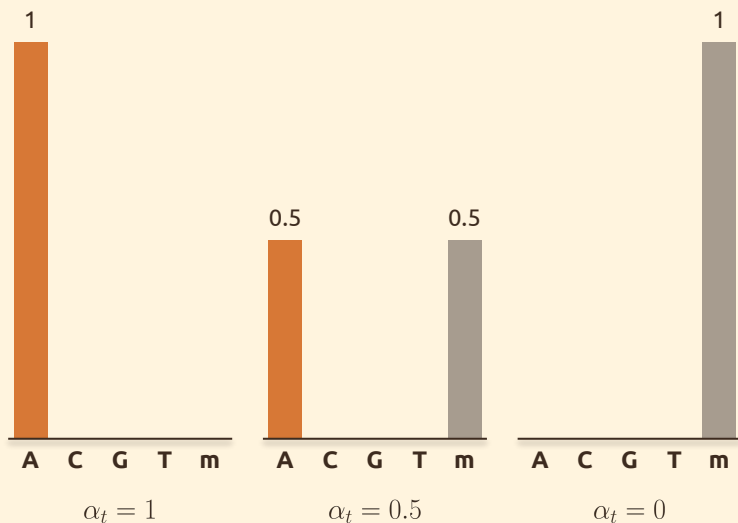
The lost probability goes to the mask, so the process ends at all masks



Under uniform noise the clean base stays with the same probability α_t

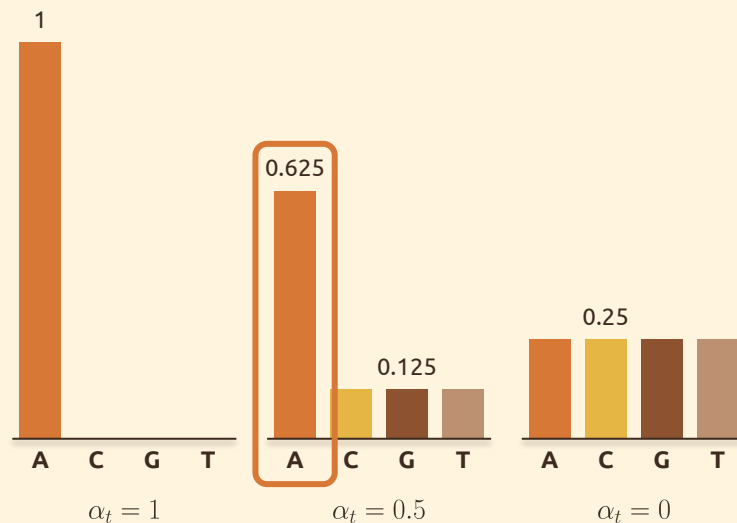
Masking: $n = \delta_m$

$$q_t(\cdot | A)$$



Uniform replacement: $n = 1/K$

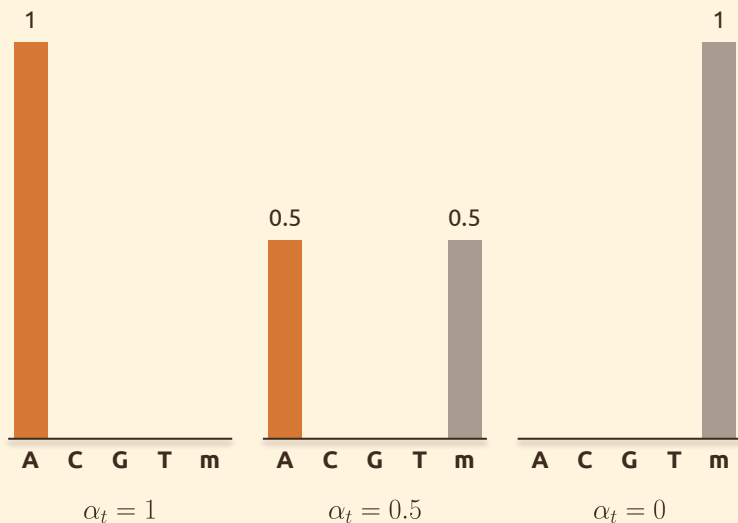
$$q_t(\cdot | A)$$



The lost probability spreads over every base, so the process ends uniform

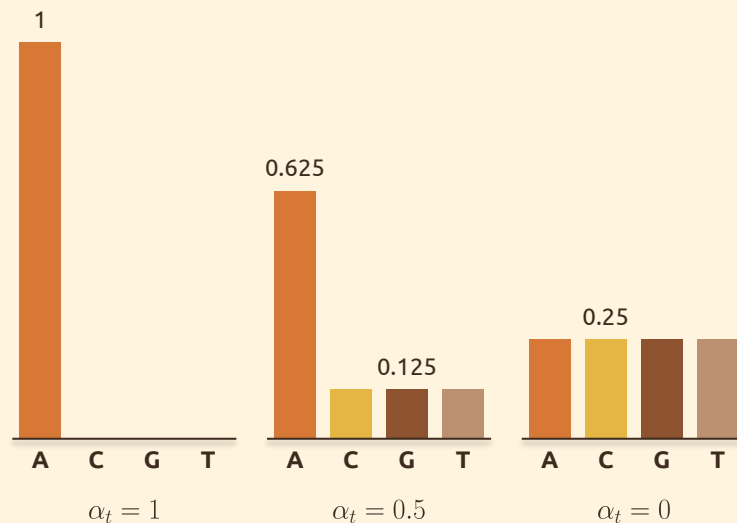
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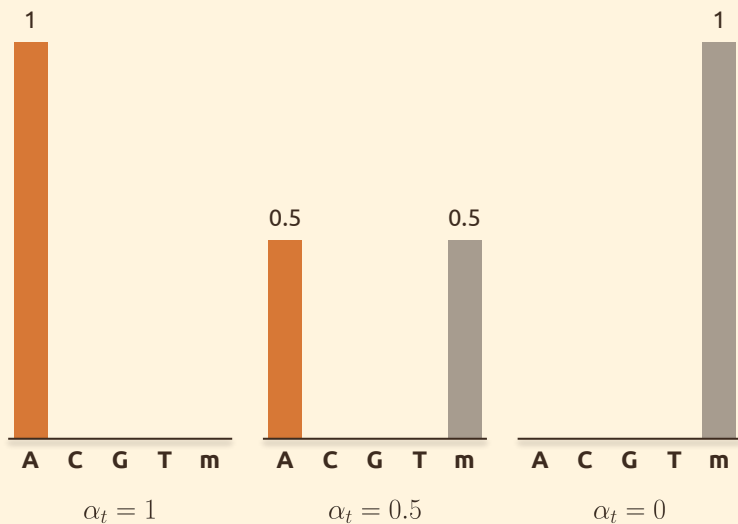
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Both kernels follow α_t , and the lost probability spreads according to n

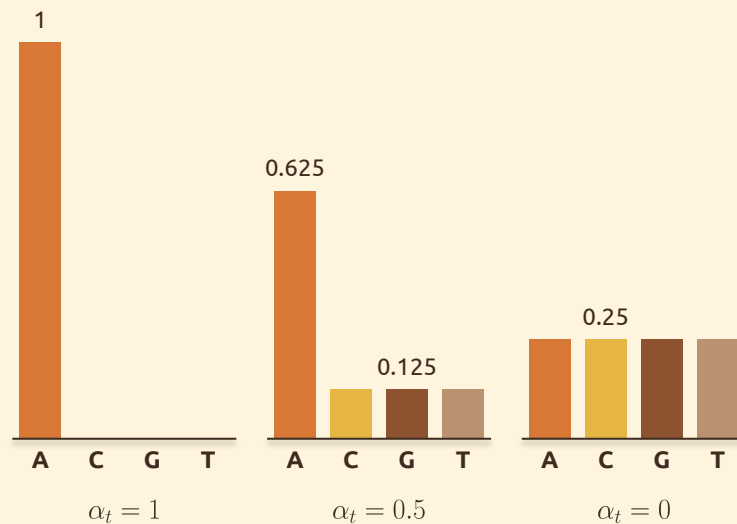
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$$e^{\tau R_{\pi}} = \frac{1}{2}I + \frac{1}{2} \cdot \frac{1}{4} \mathbf{1}\mathbf{1}^{\top} = \begin{pmatrix} 0.625 & 0.125 & 0.125 & 0.125 \\ 0.125 & 0.625 & 0.125 & 0.125 \\ 0.125 & 0.125 & 0.625 & 0.125 \\ 0.125 & 0.125 & 0.125 & 0.625 \end{pmatrix}$$

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With $\alpha = 1/2$, the matrix exponential of τ times R_{π} equals one half times the identity plus one half times one quarter times $\mathbf{1}\mathbf{1}^{\top}$, which is 0.625 on the diagonal and 0.125 everywhere else.

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A clean A stays A with probability 0.625. It survives half the time, and one quarter of the fresh draws land on A again.

Let's compute one matrix exponential for DNA

Take uniform π over A, C, G, T and accumulated rate $\tau = \log 2$, so $\alpha = e^{-\tau} = 1/2$.

$$e^{\tau R_{\pi}} = \frac{1}{2}I + \frac{1}{2} \cdot \frac{1}{4} \mathbf{1}\mathbf{1}^{\top} = \begin{pmatrix} 0.625 & 0.125 & 0.125 & 0.125 \\ 0.125 & 0.625 & 0.125 & 0.125 \\ 0.125 & 0.125 & 0.625 & 0.125 \\ 0.125 & 0.125 & 0.125 & 0.625 \end{pmatrix}$$

A clean A stays A with probability 0.625. It survives half the time, and one quarter of the fresh draws land on A again.

Split τ into n Euler steps and compare with the exponential:

n	1	2	10	∞
$(I + \frac{\tau}{n}R_{\pi})^n(A, A)$	0.480	0.570	0.616	0.625

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The entry from A to A of $(I + \tau R_{\pi}/n)^n$, the product of n Euler steps of size τ/n , is 0.480 for $n = 1$, 0.570 for $n = 2$, and 0.616 for $n = 10$, and it approaches the exact value 0.625.

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In D3PM, we multiply transition matrices step by step, and in continuous-time models, we exponentiate the rates.

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The zero direction holds the distribution we end at. Every other direction decays like $e^{-\tau}$, which is the survival probability α_τ .

The process settles in the zero direction, so let's look at it

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Since π times M equals π , the product π times R_π equals πM minus π , which is zero.

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Masking ends at the all-mask sequence, and uniform noise ends at uniformly random tokens. So generation starts from these.